DATA CENTER FIBER ROUTING -- TOWER RIDGE NODE

Drawing: DC-TR-FBR-001 Rev D | Scale: NTS | Date: 2026-04-28

DWG-DC-TR-FBR-001-D

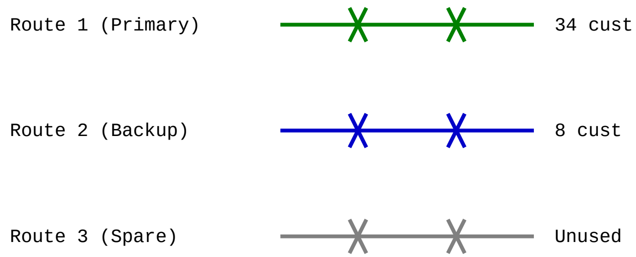
ODF PANEL A -- INCOMING

P01 ACT	P02 ACT	P03 ACT	P04 ACT
P05 ACT	P06 ACT	P07 ACT	P08 ACT
P09 ACT	P10 ACT	P11 ACT	P12 ACT
P13 ACT	P14 ACT	P15 ACT	P16 ACT
P17 ACT	P18 ACT	P19 ACT	P20 ACT
P21 ACT	P22 ACT	P23 ACT	P24 ACT
P25 ACT	P26 ACT	P27 ACT	P28 ACT
P29 ACT	P30 ACT	P31 ACT	P32 ACT
P33 ACT	P34 ACT	P35 SPA	P36 SPA
P37 SPA	P38 SPA	P39 SPA	P40 SPA
P41 SPA	P42 SPA	P43 SPA	P44 SPA
P45 SPA	P46 SPA	P47 SPA	P48 SPA

ODF PANEL B -- DISTRIBUTION

D01 OK	D02 OK	D03 OK	D04 OK
D05 OK	D06 OK	D07 OK	D08 OK
D09 OK	D10 OK	D11 OK	D12 OK
D13 OK	D14 OK	D15 OK	D16 OK
D17 OK	D18 OK	D19 OK	D20 OK
D21 OK	D22 OK	D23 OK	D24 OK
D25 OK	D26 OK	D27 OK	D28 OK
D29 OK	D30 OK	D31 RSK	D32 RSK
D33 RSK	D34 RSK	D35 RSK	D36 RSK
D37 RSK	D38 RSK	D39 RSK	D40 RSK
D41 RSK	D42 RSK	D43 ---	D44 ---
D45 ---	D46 ---	D47 ---	D48 ---

FIELD CABLE ROUTES -- TOWER RIDGE



LEGEND:

X = Splice point --- = Fiber run

ACT = Active SPA = Spare RSK = At Risk

PANEL & TRUNK UTILIZATION SUMMARY

Panel	Capacity	Active	Spare	At Risk	Failed	Utilization
ODF-A (In)	48 ports	34	14	0	0	71%
ODF-B (Dist)	48 ports	30	6	12	0	63%
Trunk 1	12 strand	10	0	1 (Str11)	0	83%
Trunk 2	12 strand	4	6	2 (Str5,7)	0	33%
Trunk 3	12 strand	0	12	0	0	0%

*CRITICAL: Route 2 shares conduit (2019, pre-standard)
with Route 1 in Sections 3-5. If conduit
fails, BOTH routes down. 42 cust, NO redundancy.
-- Nakamura, 04/28*

K. Nakamura, PE
Network Engineer
Date: 2026-04-28

--- REDUNDANCY GAP ANALYSIS ---
Route 2 merges into Route 1 conduit at Vault TV-2 (start of Section 3).
Shares conduit CDT-TR-003, CDT-TR-004, CDT-TR-005 through Section 5.
Installed 2019, before Zava redundancy standard ZNS-2021-RED-004 (adopted 2021).
Never retrofitted during 2022-2023 compliance upgrade program.

IF CONDUIT FAILS: Route 1 (34 customers) + Route 2 (8 customers) = 42 total.
D31-D42 on Panel B = 12 distribution ports serving these 42 customers.
No alternative physical path exists. Single point of failure.

IMMEDIATE: Aerial bypass for Route 2 before conduit repair (~$12K, 2-3 days)
LONG-TERM: Separate conduit for Route 2 Sections 3-5 (~$85K, Phase 2)

Refs: FSE-2026-0041 (strand degradation), PLT-DC-TR-001-E (DC termination),
AUD-2026-04-15-TR9 (field discovery at 17:00), PO-2026-0391 (materials)